

Module 1: Introduction to Performance Testing

1. What is Performance Testing?

Answer:

Performance testing evaluates how a system behaves under a particular workload in terms of **speed, stability, scalability, and responsiveness**.

2. Why is Performance Testing required?

Answer:

- To identify **bottlenecks**
 - To ensure system works under **expected load**
 - To avoid **crashes in production**
 - To validate **response time and throughput**
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3. When is Performance Testing required?

Answer:

- Before production release
 - After major code changes
 - When user load increases
 - During capacity planning
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4. What should be tested in Performance Testing?

Answer:

- Response time
 - Throughput
 - Server resource usage (CPU, memory)
 - Database performance
 - Network latency
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5. What is the objective of Performance Testing?

Answer:

- Validate system performance under load
 - Identify system limits
 - Ensure stability under stress
 - Improve user experience
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6. Define Performance Testing.

Answer:

It is a type of non-functional testing that measures system performance under varying conditions like load, stress, and scalability.

7. What is SDLC in Performance Testing?

Answer:

Performance testing is integrated into **Software Development Life Cycle (SDLC)** phases like:

- Requirement analysis
 - Design
 - Implementation
 - Testing
 - Deployment
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8. What is Performance Testing Life Cycle (PTLC)?

Answer:

Steps include:

1. Requirement gathering
 2. Test planning
 3. Script development
 4. Test execution
 5. Result analysis
 6. Reporting
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9. What is Architecture in Performance Testing?

Answer:

It refers to system design including:

- Web server
- App server
- Database
- Load balancers

Understanding architecture helps find bottlenecks.

10. What are Environments in Performance Testing?

Answer:

- Dev
- QA
- Staging
- Production-like

A **production-like environment** is preferred for accurate results.

11. What are types of Performance Testing?

Answer:

- Load Testing
 - Stress Testing
 - Spike Testing
 - Endurance Testing
 - Volume Testing
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12. What are roles and responsibilities in Performance Testing?

Answer:

- **Performance Tester:** Script creation, execution
 - **Test Lead:** Strategy & planning
 - **Developers:** Fix performance issues
 - **DBA:** Database tuning
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◆ Module 2: NFR (Non-Functional Requirements)

13. What is NFR in Performance Testing?

Answer:

Non-functional requirements define **performance expectations** like:

- Response time < 2 sec
 - 1000 concurrent users
 - 99% availability
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14. What are Performance Testing Requirements?

Answer:

- Expected user load
 - Peak load
 - Response time criteria
 - Throughput expectations
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15. What is a Performance Test Strategy?

Answer:

A document that defines:

- Scope of testing
 - Test types
 - Tools used
 - Workload model
 - Risks
-

◆ Module 3: JMeter Setup

16. What is JMeter?

Answer:

Apache JMeter is an open-source tool used for **performance and load testing of web applications**.

17. Why is classpath setup required in JMeter?

Answer:

To include:

- External libraries
 - JDBC drivers
 - Plugins
-

18. Why do we increase TCP/IP ports?

Answer:

To avoid **port exhaustion** when simulating many users.

19. What is port exhaustion?

Answer:

When system runs out of available network ports due to high number of connections.

20. Why increase JMeter heap memory?

Answer:

To handle large tests and avoid:

- Memory errors
 - Crashes
-

◆ Module 4: JMeter Recording

21. What is Proxy setup in JMeter?

Answer:

It captures browser requests using **HTTP(S) Test Script Recorder**.

22. What is JMeter Recording?

Answer:

Process of capturing real user actions and converting them into test scripts.

23. What is HAR to JMX?

Answer:

Converting **HAR file (browser recording)** into JMeter script (.jmx).

24. What is BlazeMeter in recording?

Answer:

A cloud-based tool used to:

- Record scripts
 - Convert to JMeter format
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25. What is Playback in JMeter?

Answer:

Running recorded scripts to simulate user behavior.

26. How do you validate requests and responses?

Answer:

Using:

- Response Assertions
 - Regular Expressions
 - JSON Extractors
-

27. What is correlation in recording?

Answer:

Handling dynamic values (like session IDs) so scripts run correctly.

28. What is request and response validation?

Answer:

- Request: Verify correct data is sent
- Response: Verify correct data is received

💡 Bonus Interview Questions

29. Difference between Load Testing and Stress Testing?

Answer:

- Load Testing → normal expected load
- Stress Testing → beyond limits

30. What is throughput?

Answer:

Number of requests processed per second.

31. What is response time?

Answer:

Time taken by server to respond to a request.

32. What is concurrency?

Answer:

Number of users accessing system at same time.

1. What is the primary goal of performance testing?

- A. Find functional bugs
- B. Check UI design
- C. Evaluate system speed and stability
- D. Test security

2. Which of the following is NOT a performance metric?

- A. Response Time
 - B. Throughput
 - C. Test Cases Passed
 - D. CPU Usage
-

3. Performance testing is mainly a:

- A. Functional testing
 - B. Non-functional testing
 - C. Unit testing
 - D. Regression testing
-

4. What does response time measure?

- A. Time to send request
 - B. Time to receive response
 - C. Total time between request and response
 - D. Server uptime
-

5. Which testing checks system behavior beyond limits?

- A. Load Testing
 - B. Stress Testing
 - C. Smoke Testing
 - D. Regression Testing
-

6. Which environment is best for performance testing?

- A. Development
 - B. Testing
 - C. Production-like
 - D. UAT only
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7. Throughput means:

- A. Number of users
- B. Number of requests per second
- C. Server memory
- D. Test cases executed

8. What is scalability testing?

- A. Testing UI
 - B. Testing database
 - C. Testing system growth capability
 - D. Testing login functionality
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-

◆ Module 2: NFR & Strategy

9. NFR stands for:

- A. Non-Functional Requirement
 - B. New Functional Requirement
 - C. Network Failure Response
 - D. None
-

10. Which is an example of NFR?

- A. Login functionality
 - B. Add to cart
 - C. Response time < 2 seconds
 - D. Logout feature
-

11. Performance test strategy defines:

- A. Code logic
 - B. UI design
 - C. Testing approach and scope
 - D. Database schema
-

12. Peak load means:

- A. Minimum users
- B. Average users
- C. Maximum expected users
- D. Zero users

◆ Module 3: JMeter Setup

13. Apache JMeter is used for:

- A. UI testing
 - B. Performance testing
 - C. Security testing
 - D. Manual testing
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14. Why increase TCP/IP ports in JMeter testing?

- A. Improve UI
 - B. Avoid port exhaustion
 - C. Increase memory
 - D. Faster coding
-

15. Heap memory in JMeter is increased to:

- A. Improve UI
 - B. Handle large load
 - C. Reduce test cases
 - D. Increase network speed
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16. What is port exhaustion?

- A. CPU overload
 - B. Memory crash
 - C. No available network ports
 - D. Disk failure
-
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◆ Module 4: JMeter Recording

17. HTTP(S) Test Script Recorder is used for:

- A. Execution
 - B. Reporting
 - C. Recording scripts
 - D. Debugging
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18. Proxy in JMeter is used to:

- A. Increase speed
 - B. Capture requests
 - C. Store data
 - D. Run scripts
-

19. HAR file is:

- A. Java file
 - B. Log file
 - C. Browser network recording
 - D. Script file
-

20. What is correlation?

- A. UI design
 - B. Handling dynamic values
 - C. Writing test cases
 - D. Running scripts
-

21. Which is used to validate response in JMeter?

- A. Timer
 - B. Listener
 - C. Assertion
 - D. Controller
-

22. Playback means:

- A. Recording again
- B. Running recorded script
- C. Saving script
- D. Deleting script

◆ Bonus MCQs (Important for Interviews)

23. Concurrency means:

- A. Sequential execution
 - B. One user at a time
 - C. Multiple users at same time
 - D. No users
-

24. Which test checks system for long duration?

- A. Load Testing
 - B. Endurance Testing
 - C. Spike Testing
 - D. Smoke Testing
-

25. Spike testing means:

- A. Gradual increase
 - B. Sudden increase in load
 - C. Decrease load
 - D. No load
-
-

✓ Answers

- 1. C
- 2. C
- 3. B
- 4. C
- 5. B
- 6. C
- 7. B
- 8. C
- 9. A
- 10. C
- 11. C
- 12. C

- 13. B
- 14. B
- 15. B
- 16. C
- 17. C
- 18. B
- 19. C
- 20. B
- 21. C
- 22. B
- 23. C
- 24. B
- 25. B